Tanmay Banjari

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.85
FIITJEE Miyapur, Hyderabad	Class XII (TSBIE)	2023	95.50%
Phoenix Greens School, Kokapet, Hyderabad	Class X (CBSE)	2021	97.00%

WORK EXPERIENCE

Razorpay Software Limited | *Software Engineer Intern*

- Built **Kubernetes autoscaling** for a Kafka worker, allowing it to scale from **2 to 10 pods** based on CPU usage and message backlog, with capacity increasing by **up to 100% every 30s** and gradual scale-down during low traffic
- Added **Redis caching** for terminal details across **critical payment flows**, **eliminating an external service call on cache hits** and reducing **repeated Terminal Service requests** with safe fallback on cache failures.
- Fixed HTTP timeout issues across **critical payment dependencies**, changing potentially **indefinite waits to bounded timeouts (400 ms-30 s)** and adding **circuit breakers with 100-request concurrency limits** to prevent downstream failures from blocking the payment service
- Built a **persistent failure-tracking system** for recurring payments that retained account-status failures across cooldowns and **blocked payment creation after 2 failures**, preventing repeated debit attempts and unnecessary downstream gateway calls

PROJECTS

IoT Delivery Fleet Monitoring Gateway | *C++, epoll, Systems Programming, Python* | Link

- **Built a high performance C++ telemetry gateway** using **Linux epoll** and **non blocking sockets** with SO REUSEPORT worker loops to ingest high volume vehicle updates without per operation mutex locks
- **Implemented per IP token bucket rate limiting** and worker thread pools to validate incoming telemetry JSON payloads and protect system ingestion from traffic spikes
- **Engineered a durable memory mapped append only log** for crash resilient event storage alongside an **in memory ring buffer** for constant time state queries and live fleet dashboard rendering
- **Validated network performance and streaming recovery** through **chaos testing** using abrupt process termination, verifying **at least once delivery**, consumer offset reconnects, and **50000 concurrent TCP connections**

Accelerated kNN and Range Search using Ball Trees | *C++, Python, Data Structures* | Link

- Built Ball Trees using two partitioning strategies, implementing **median split partitioning in C++** and **PCA based projection in Python** to index multidimensional data into bounding spheres
- Implemented **kNN and range queries** using **branch and bound pruning** with a **max heap priority queue**, skipping irrelevant subtrees to outperform **brute force linear search**
- Benchmarked tree performance against naive search using a **custom C++ testbench**, evaluating **construction time, tree depth, and node traversals** on synthetic datasets like **Moons, Swiss Roll**, and **MNIST image vectors**

Resilient Urban Air Traffic Control System | *Python, Multi-Agent Systems, Distributed Algorithms* | Link

- **Engineered a two-tier architecture** featuring a centralized **Drone Traffic Controller (DTC)** for macro routing, corridor allocation, and capacity management across a 60-agent heterogeneous fleet
- **Implemented a priority arbitration engine** (**Emergency > eVTOL > Delivery**) paired with boids formation rules and **ADMM distributed optimization** to resolve high-density merge conflicts
- **Designed a resilient defense pipeline** using **W-MSR consensus** to filter falsified state broadcasts, enabling graceful degradation during DTC failure while enforcing **STL safety bounds**

Bayesian Trading Strategy Framework | *Python, Quantitative Finance, Statistics* | Link

- Built a **Bayesian strategy selection framework in Python**, replacing heuristic backtesting with probabilistic estimates of strategy win rates and expected returns
- Implemented **Beta Binomial and Normal Normal conjugate models** with profit weighted sequential updates to jointly model trade win probability and return magnitude
- Backtested on **2500 days of Bharti Airtel data**, achieving **78% cumulative return for RSI vs 24% for EMA**, and validated strategy edge using **posterior predictive checks and forward testing** on unseen data

SKILLS

Languages: C | C++ | Python | SQL | Go (basic)

Tools and Libraries: Git | Latex | Numpy | Pandas | Matplotlib | Scikit-learn | Docker

RELEVANT COURSES

Computer Science: Data Structures & Algorithms, Approximation Algorithms, Database Management Systems, Operating Systems

Artificial Intelligence: Foundations of Machine Learning, Natural Language Processing, Multi-Agent Systems

Mathematics: Number Theory, Combinatorics, Spectral Graph Theory, Probability Theory, Linear Algebra, Bayesian Statistics, Predictive Modelling

Miscellaneous: Business Analytics Using Python, Introduction to Entrepreneurship, Communication Skills

EXTRACURRICULARS

Competitive Programming: *1600+* rated in CodeChef, solved over *650+* problems in Codeforces, *300+* problems in LeetCode.

Elan & nVision | *Annual Techno-Cultural Fest, IIT Hyderabad* | Head (Infra & Security), Coordinator (Production)

Class Representative (4th Year) | *Department of Mathematics & Computing, IIT Hyderabad*

Sports: Basketball, Table Tennis, Volleyball, Cricket, Badminton